



QUANTIUM RETAIL ANALYSIS



CHIPS SALES ANALYSIS

About Quantum:

Quantum is a leading global data analytics company that helps retailers better understand their customers and sales patterns. Their goal is to drive business growth, improve decision making, and reduce risks through data driven strategies.

Objective of the Analysis:

The objective of this analysis is to understand the key drivers of chips sales and customer purchasing behavior across various chips products in order to provide actionable insights and recommendations.

Overview of the Analysis :

This analysis focuses on the chips product category, examining periodic sales trends and customer behavior. It draws insights from key features such as: Loyalty Card Number , Life Stage , Premium Customer, Store Number , Transaction ID , Product Number , Product Name , Product Quantity and Total Sales.



KEY FINDINGS

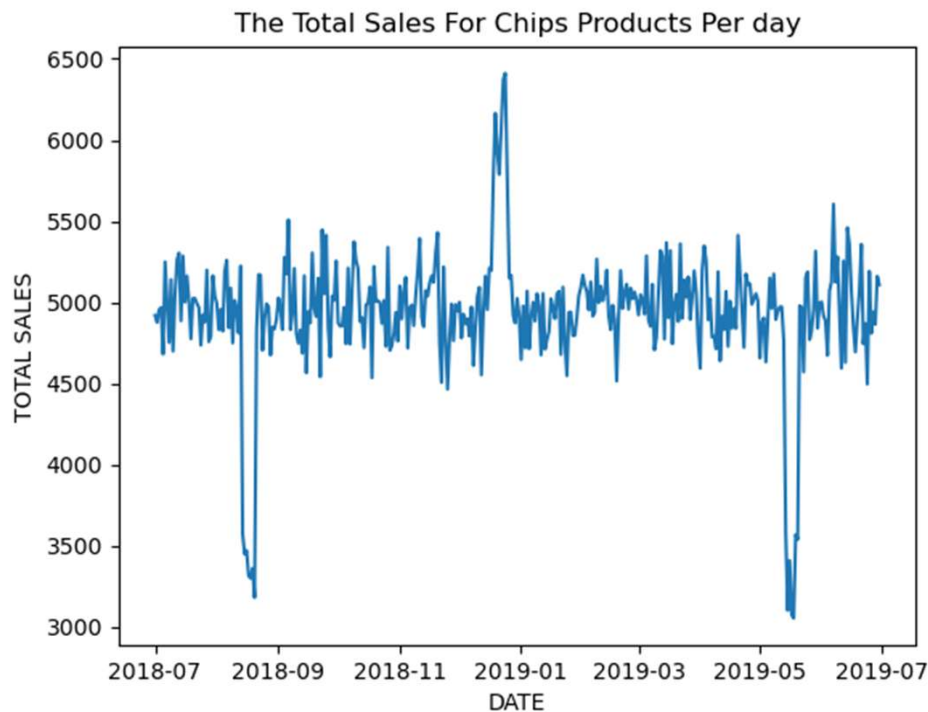
Task 1: Chips insights

- The high and low performing months of chips sales
- The possible causes of drop in sales
- The drivers of consumer purchasing behavior
- The impact of loyalty card number on sales
- The most purchased chips brands
- The influence of unit price on consumer purchasing behavior
- The product size consumers admire most and those that consumers admire less
- The relationship between life stage of customers and their purchasing preferences

Task 2: Control and Trial Stores

- Among the trial and control stores, the most highly sales store
- The kind of strategies to implement in store regions near to the various trial and control stores to target customers
- Using the trial store with high performing sales strategies to conduct A/B test
- Identifying and maintaining chips brands admired by most customers based on the store location
- Determining the significant differences between control and trial stores

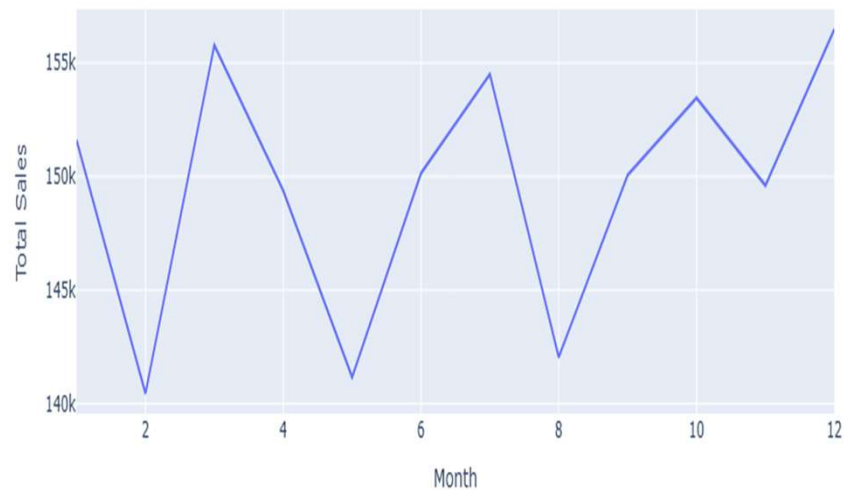
Total sales performance of chips per day



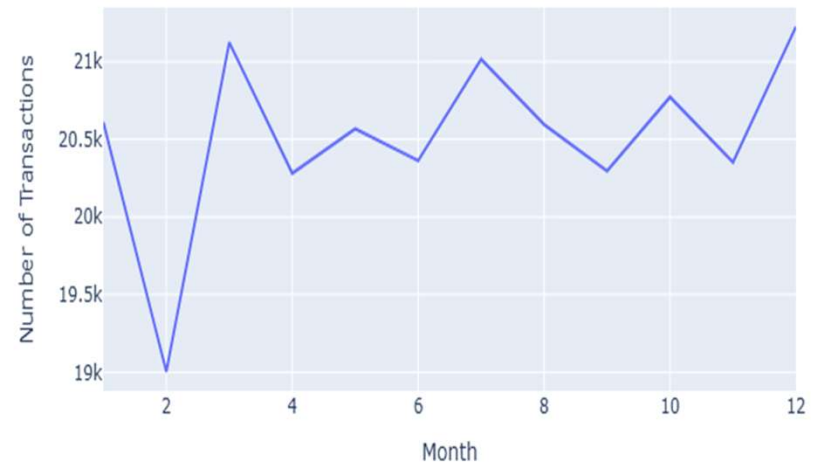
- It's seen that in the month of December close to the beginning of the month of January there's a rise in product sales indicating a potential boosting of chips sales during December. This rise in sales is due to the holyday period where a lot of people goes for holiday and purchases a lot of products.
- Retailers can use this period as an advantage to promote and advertise low sales products to attract more customers
- There is a steep decline in sales during the month of May 2018 and August 2019 and this may be due to the incidence of not opening the stores early . Retailers need to pay attention to this period to possibly identify the causes of the steep drop in sales during these period.

Total monthly sales and transactions of chips for low and high performing months

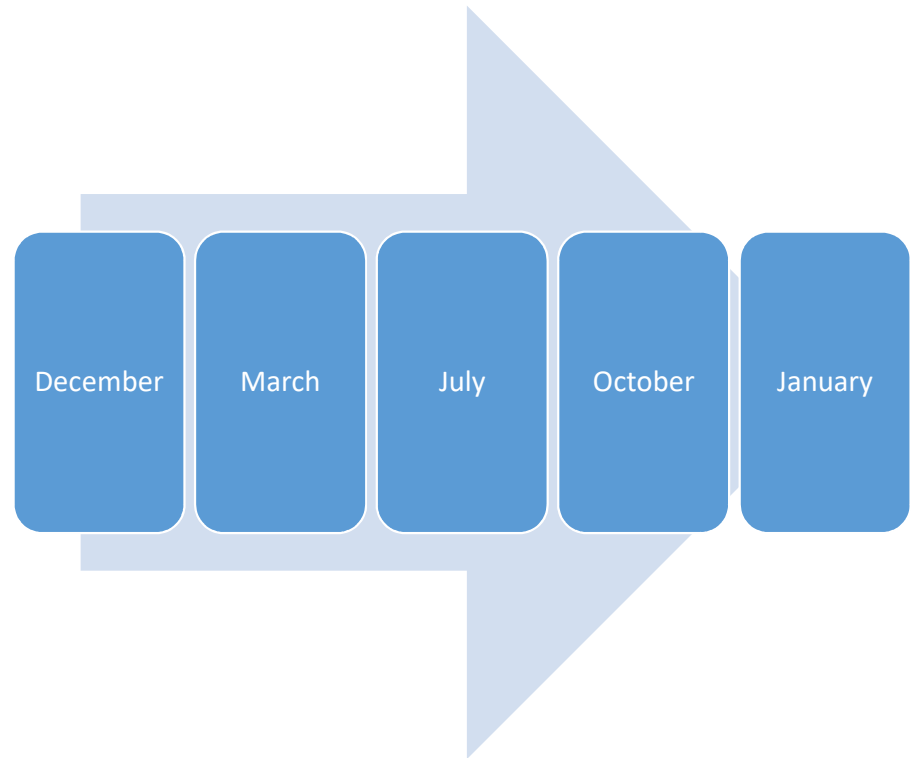
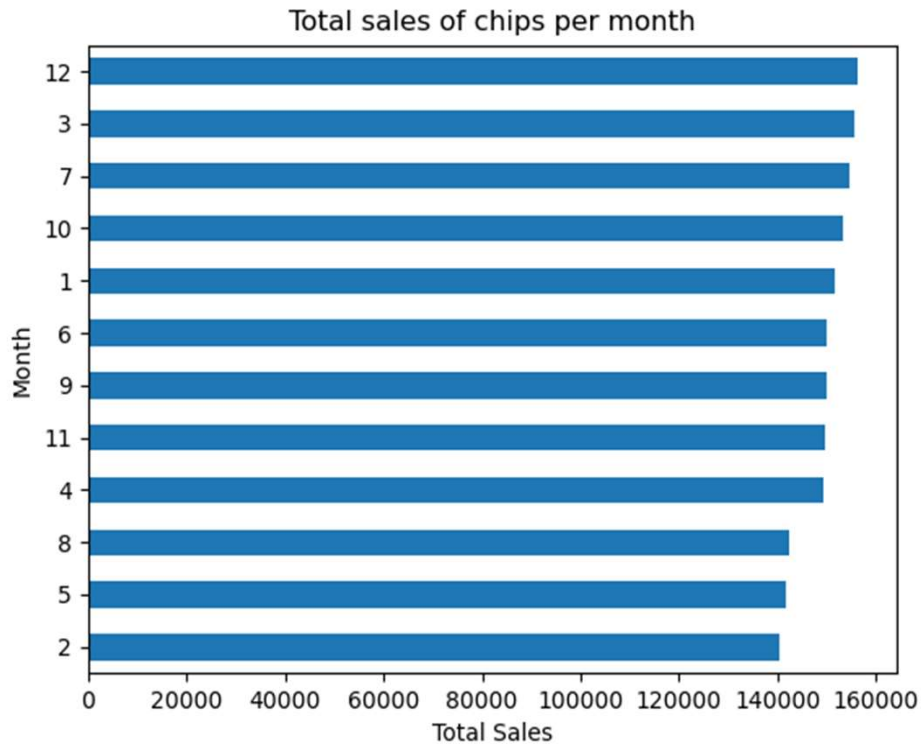
Total Monthly Sales



Monthly Transactions

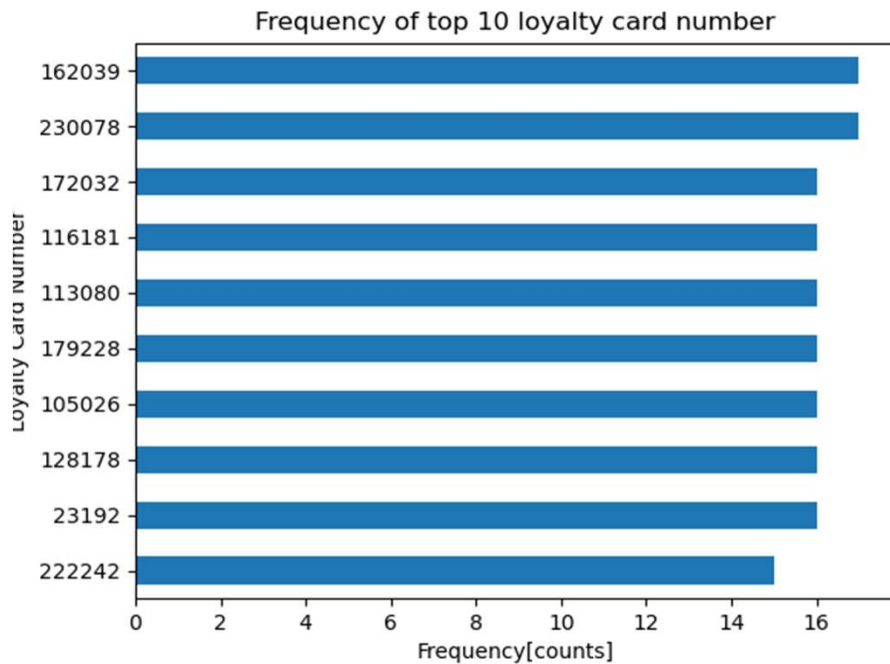


Total monthly sales of chips products



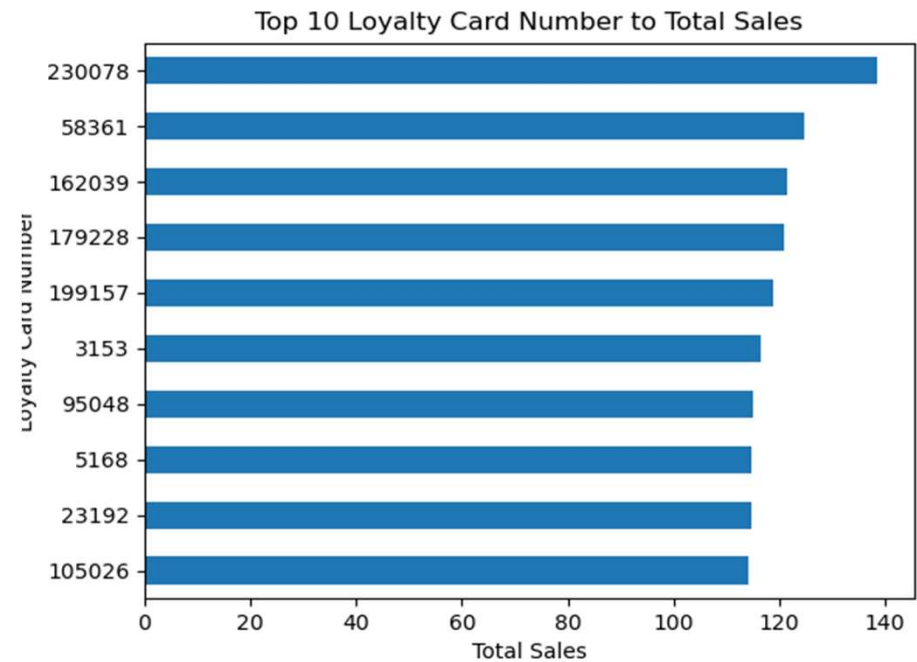
The Drivers of consumer purchasing behavior

This indicate the impact of given out loyalty card to customers, which leads to the increase of sales



Top sales by loyalty card number to individual customer.

This indicates that customers with more loyalty card number tend to purchase more of the products

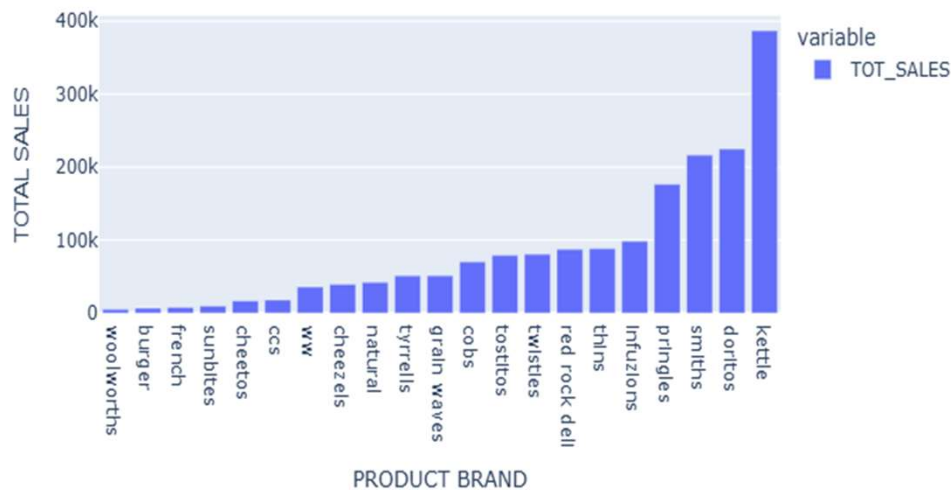


The brand of chips preferred by customers

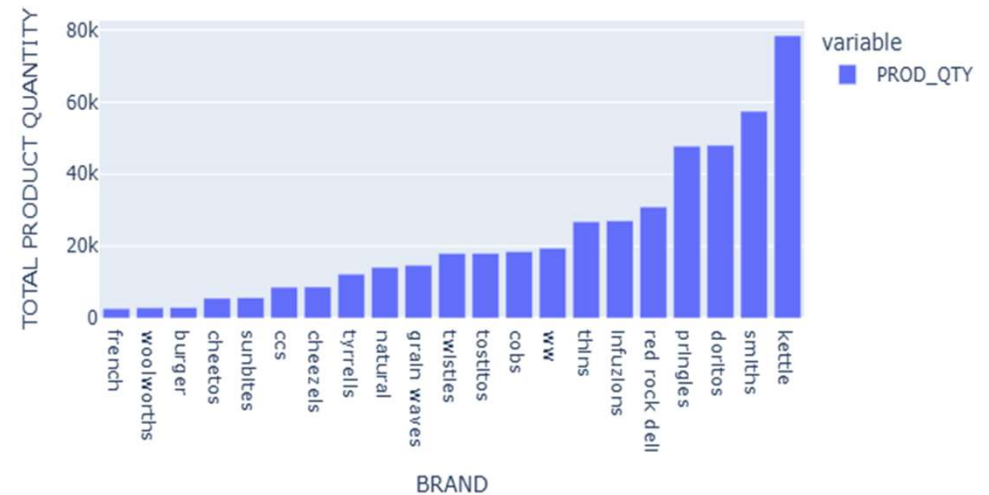


Decreasing order of chips preference by customers . This shows the quantity of chips purchased by customers, which will help retailers to maintain them and to improve sales.

PRODUCT BRAND BY TOTAL SALES



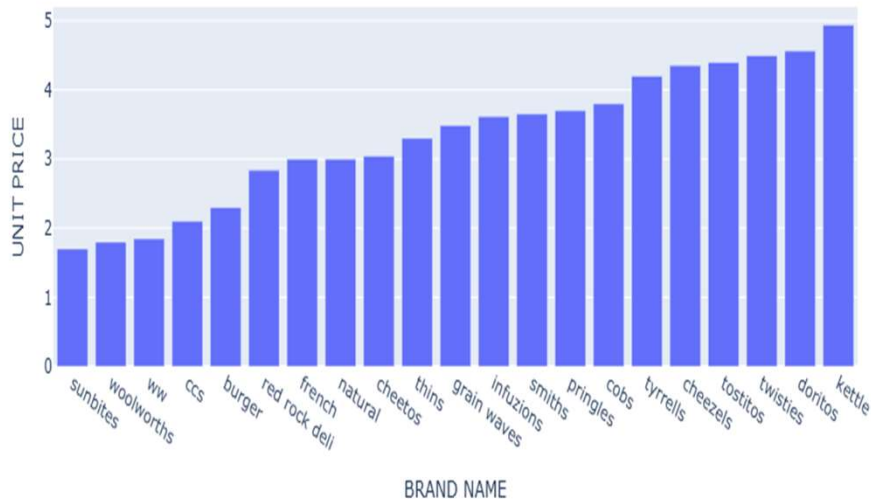
TOTAL QUANTITY OF BRAND PURCHASED



The impact of unit price on consumer purchasing behavior

Wealthy customers tend to buy chips products with high unit price due to it's quality and taste

BRAND PER UNIT PRICE

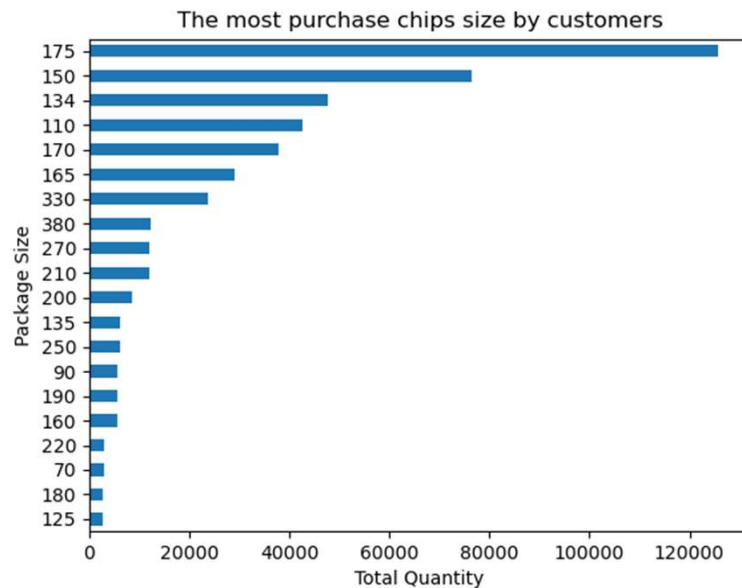


The key insights

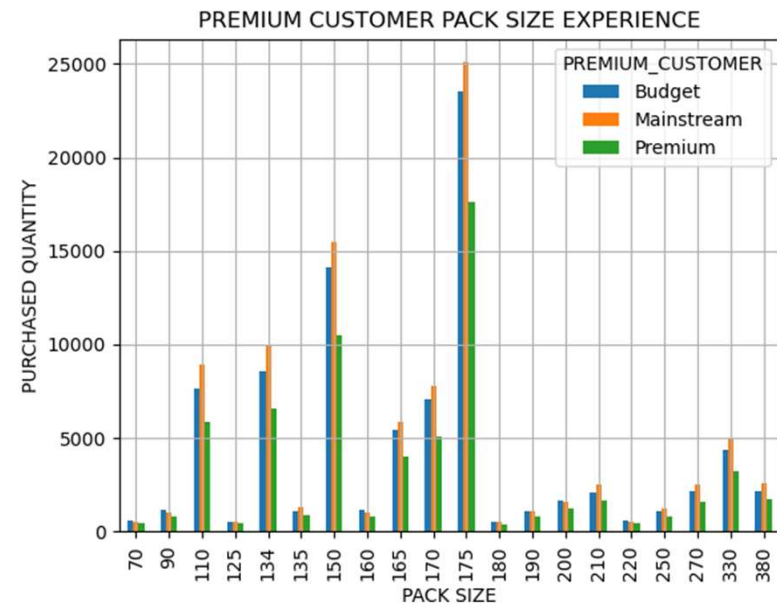
- Chips products such as kettle, doritos, twisties, tostitos, cheezels and tyrells in decreasing order of unit price are mostly purchased by wealthy customers due to it's quality.
- Cobs, pringles, smiths, infuzions, grain waves, thins, cheetos, natural and French chips products are usually purchased by averaged customers.
- Lastly, chips brand such as red rock deli, burger, ccs, woolworths and sunbites are usually purchased by budgeted customers due to it's low average unit price.
- Therefore targeting customers based on the unit price will improve sales of chips. This always depends on the location of the store whether in rural or urban areas.

The preferred package size by customers

The pack size that customers purchase most.

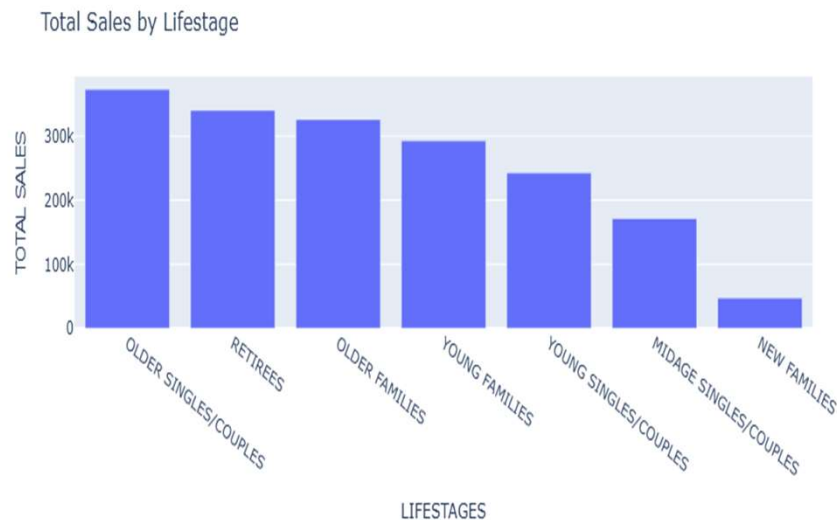


The various consumer segments with the highly preferred packed size



The various life stages with the total amount of chips products purchased

Identifying and knowing the life stages of most of the customers who contribute much to chips products. This will help retailers to maintain and target those customers to boost sales



Key insights

- The visualization shows that, generally higher purchase of chips products are coming from the following life stages of customers, older singles/couples, retirees and older families and knowing these will aid retailers to maintain those customers by given out loyalty card number.
- Also, life stages such as young families, young singles/couples, midage singles/couples and new families contributes low to total sales in a decreasing order and this maybe due to their population within the region and providing them with loyalty card number too will encourage them to purchase more goods
- The influence of these life stages on total sales can be due to their population within the region.

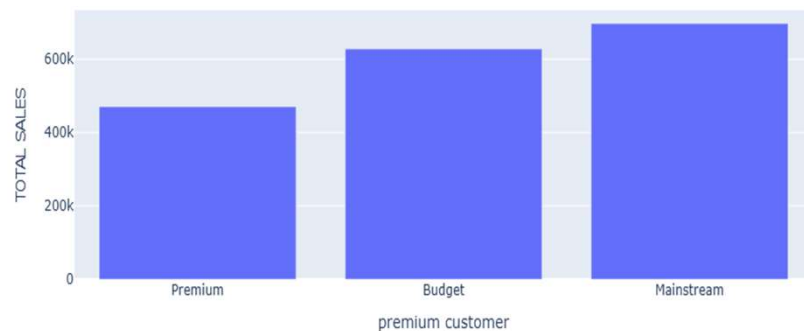
The total sales of chips products by consumer segment



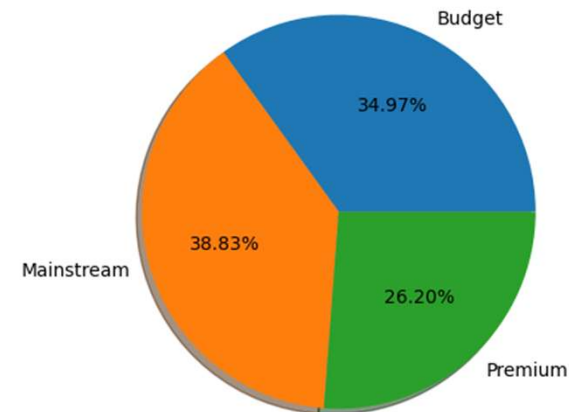
The increasing hierarchical order of contributors to high total sales.

Percentages of consumer segments on total sales

Total Sales by premium customer

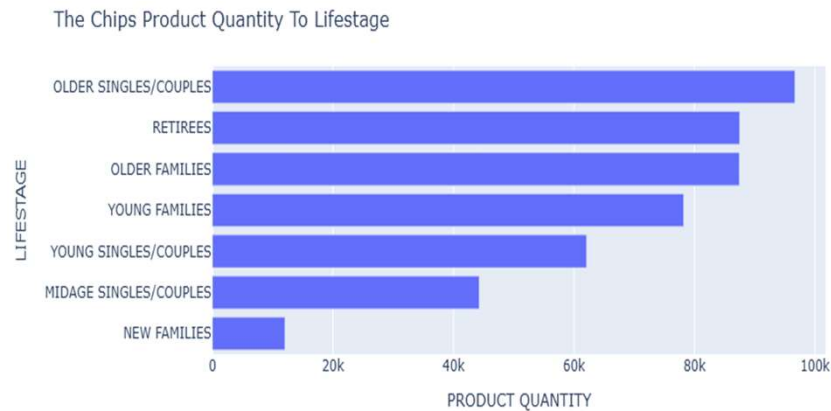


Percentages of Various Premium Consumer



The total product quantity by life stage and by customer segment

Product Quantity by Life stage



Product Quantity by Consumer Segment



The relationship between customer life stage and consumer segment

Percentage of Premium Customer by Lifestage



- Most of the older families tends to purchase low priced chips products due to the family size and expenses.
- Most of the retirees and young singles/couples tends to purchase averagely priced chips products contributing to be the highest averaged customers.
- Older singles/couples tends to purchase much of the highly priced chips products due to their financial abilities and low family expenditure, they tend to purchase chips with higher quality.
- Generally, very few of them cut across are from the new families due to low population of them within the region
- Hence, targeting these customers based on their life stage and the segment they fall under will aid boost sales.

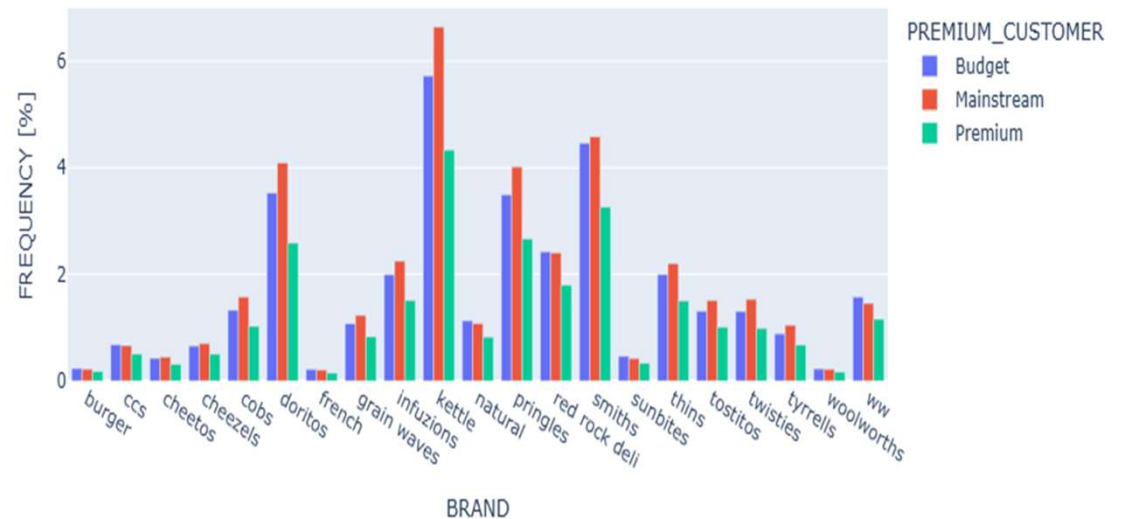
The brands of chips products preferred by the various consumer segments

Generally customers under the mainstream category tends to purchase much of the various brand of chips products. This percentage of preferences is due to their population within the region.

Budget customers tends to purchase chips such as burger, ccs, french, natural, sunbites and ww chips brand due to their lower prices.

Knowing the various segment preference will help in strategic improvement in the category sales and customer purchasing behavior.

Percentage of Brand Preference by Premium_Customer



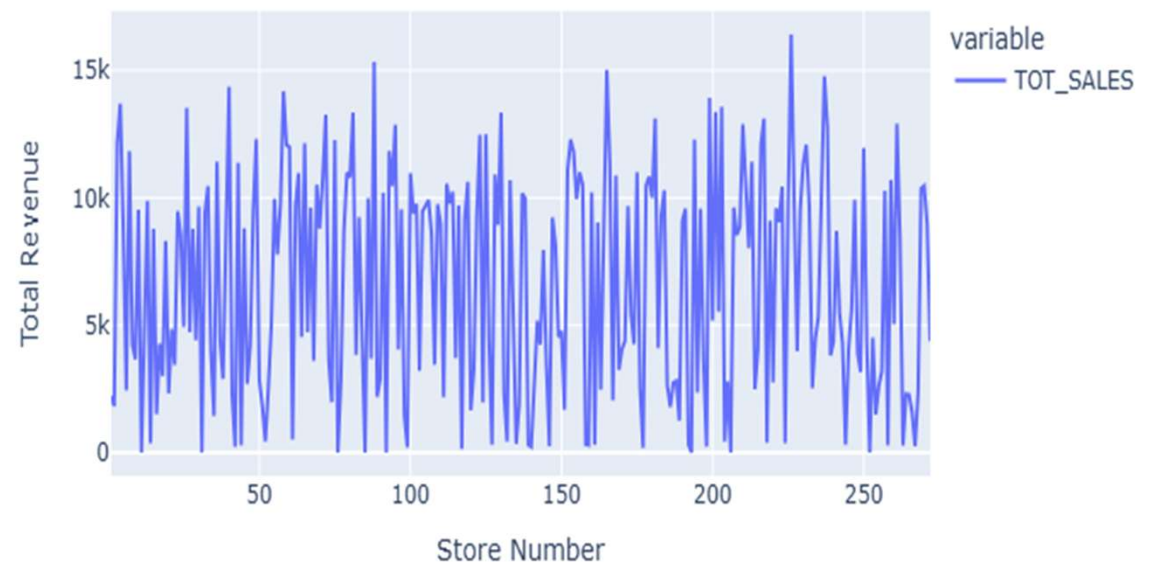
TASK 2

Both the control and trial store,
the one with high sales
performance

From the total sales of control and trial stores, it shows that trial store 88 outperforms the rest of the control and trial stores.

Hence implementing the strategic planning of trial store 88 to 50% of the stores as an A/B test and run the experiment for 3 months can help boost product sales.

Total Revenue Per Store

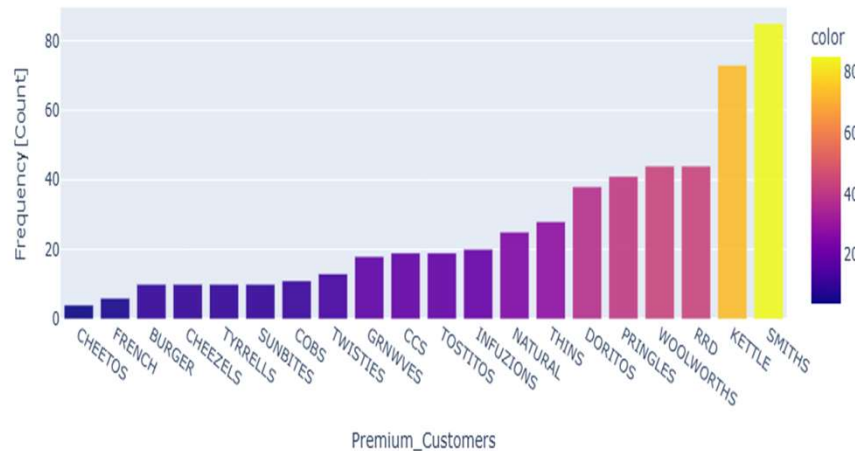


Brands preferred based on store region and their customer life stage

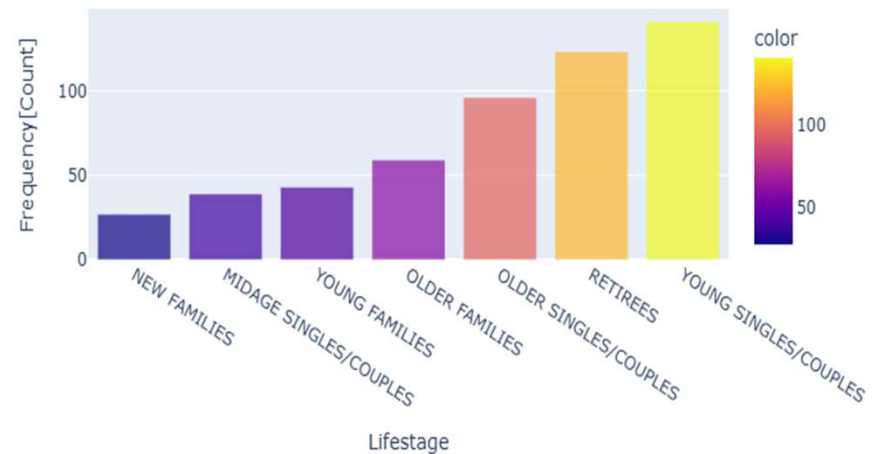
Trial store 77 : most of the customers within that region prefers more of smiths and kettle brand in a decreasing order respectively

Trial store 77: most of the customers within that region are from young singles/couples, retirees and older singles/couples. Targeting those customers within the region can boost sales.

Number of Various Brands to Trial Store 77



Number of Various Lifestages to Trial Store 77

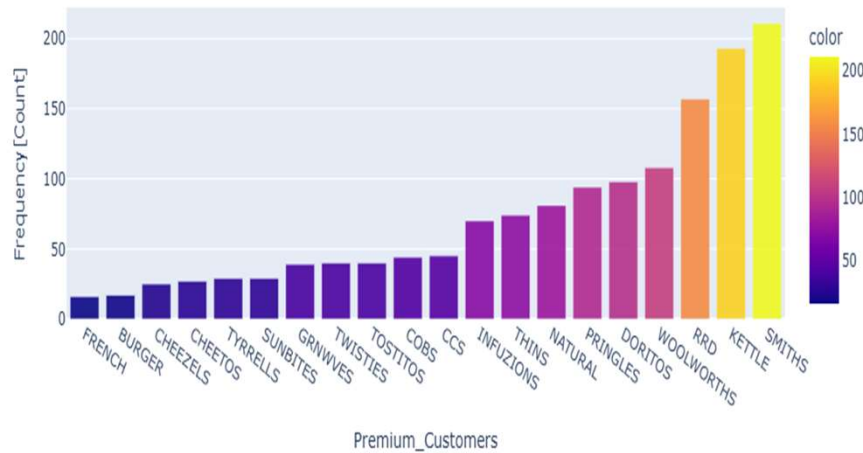


Brands preferred based on store region and their customer life stage

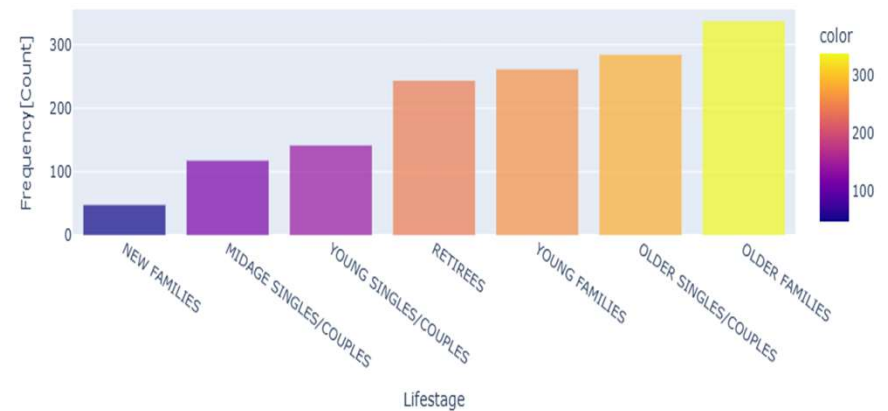
Trial store 86: most of the customers within that region prefers more of smiths, kettle and red rock deli chips products.

Trial store 86: most of the customers within that region are from older families, older singles/couples, young families and retirees. Targeting those customers within the region can boost sales.

Number of Various Brands to Trial Store 86



Number of Various Lifestages to Trial Store 86

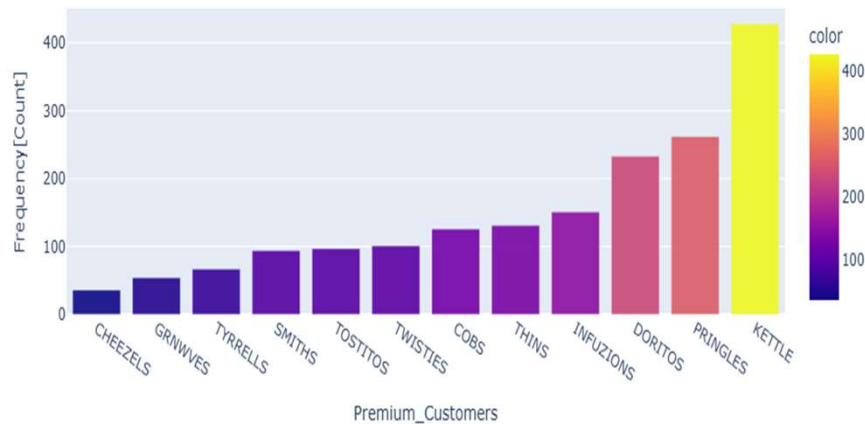


Brands preferred based on store region and their customer life stage

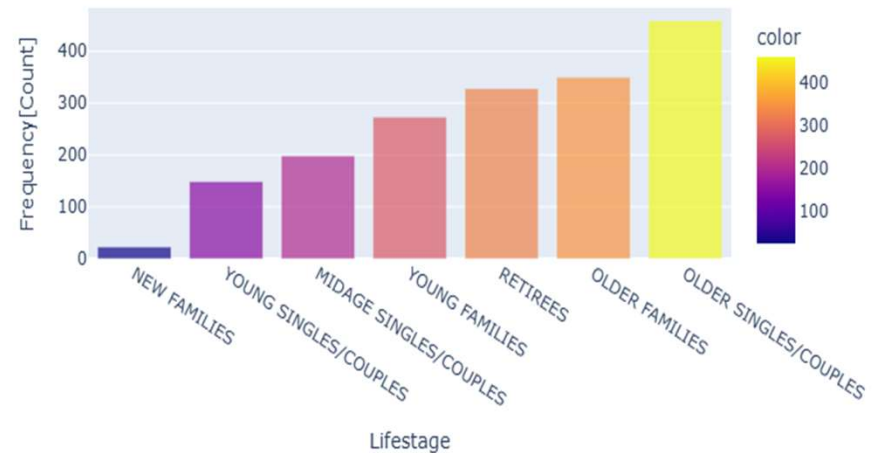
Trial store 88: : most of the customers within that region prefers more of kettle, pringles and doritos chips products.

Trial store 88:most of the customers within that region are from older singles/couples, older families and retirees. Targeting those customers within the region can boost sales.

Number of Various Brands to Trial Store 88



Number of Various Lifestages to Trial Store 88



General summary

- 1. Address Low Sales Periods:** Implement targeted promotions, events, and customer engagement activities in February, May, and August to counter seasonal sales dips.
- 2. Run Strategic A/B Testing:** Test store strategies by replicating trial store 88 approach in 50% of similar stores for 3 months, comparing performance with the other half to find the most effective method.
- 3. Targeted Customer Segmentation:** Focus marketing on high purchasing groups older singles/couples, retirees, older families, and young families while creating special offers for underperforming groups.
- 4. Maximize Promotions in Peak Months:** Intensify marketing campaigns, in store displays, and product promotions during high sales months such as March, July, October, December to leverage natural spikes in demand groups like Young Singles/Couples and New Families.
- 5. Promote Low Performing Brands in Peak Seasons:** Boost visibility for low selling brands french, woolworths, burger, cheetos, sunbites, cheezels and css during peak demand months to lift their sales.
- 6. Optimize Shelf Placement for Underperformers:** Conduct A/B testing on different shelf placements and store locations for low performing brands to determine arrangements that drive the highest sales.
- 7. Leverage Loyalty Programs:** Actively issue and promote loyalty cards to encourage repeat purchases, as high loyalty customers tend to buy more.
- 8. Boost Advertising for Popular Brands:** Increase advertising and promotional visibility for high performing brands like kettle, smiths, doritos, and pringles to sustain and grow their market share.
- 9. Enhance Store Management During Slow Periods:** Test the effect of reshuffling store managers versus keeping current ones, particularly in slow months, to identify leadership strategies that improve sales.
- 10. Monitor and Adapt to External Factors:** Track economic conditions, climate changes, and seasonal temperature patterns to dynamically adjust pricing, promotions, and product availability in response to shifts in customer buying behavior.

Expected Outcomes

- Higher seasonal sales peaks.
- Reduced sales dips in low months.
- Increase sales in stores with lower sales and to improve those with higher sales.
- Increased loyalty card engagement and brand visibility.
- Better targeting of profitable customer segments and understanding customer purchasing behavior.
- Data driven and decision making for continuous improvement.